Early Prediction of Preeclampsia Using Multi-Source Maternal Health Data

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Abstract

Preeclampsia is a life-threatening hypertensive disorder of pregnancy that often goes undetected until late-stage symptoms emerge. This paper presents a machine learning framework for early risk prediction evaluated independently across three maternal health datasets. Eight classifiers—Logistic Regression, Decision Tree, Random Forest, SVM, KNN, Naive Bayes, Gradient Boosting, and XGBoost—were trained and compared under both standard conditions and deliberately introduced adversarial scenarios, including class imbalance (SMOTE), label noise injection, and feature distribution drift, to assess real-world robustness. Each dataset then underwent a model improvement phase involving hyperparameter tuning, feature selection, and ensemble construction. Across all experiments, Random Forest proved the most consistently reliable model, achieving 84.7% test accuracy and AUC of 0.951 on the largest dataset. On the smallest and most challenging dataset, a soft-voting ensemble of three diverse classifiers reached a validation F1-score of 0.735. A key finding is that adversarial conditions affect models very differently: ensemble methods suffer under label noise while simpler probabilistic models remain stable, yet ensembles recover strongly under feature drift. These results suggest that selecting a model for clinical deployment cannot rely on benchmark accuracy alone; robustness testing under realistic data conditions is equally important.

1 Introduction

Preeclampsia is one of the leading causes of maternal and perinatal mortality worldwide, characterised by elevated blood pressure and organ damage typically appearing after the twentieth week of pregnancy. Early identification of at-risk patients is critical, as timely clinical intervention can significantly reduce complications. Despite this, diagnosis is often delayed until symptoms become severe, limiting the window for preventive care.

Traditional prediction methods rely on statistical models and clinical observations that struggle to capture the complex, nonlinear relationships among the many physiological, demographic, and biochemical factors involved in preeclampsia risk. Machine learning offers a compelling alternative, enabling models to learn from heterogeneous, high-dimensional data and uncover patterns that conventional approaches cannot detect.

Much of the existing literature evaluates models under idealised conditions using a single dataset, which does not reflect the variability encountered in real clinical settings. Data collected across different hospitals, devices, or patient populations often contains noise, labelling errors, and distributional differences that can cause even high-performing models to fail in practice. This gap between benchmark performance and real-world applicability motivates the present study.

We address this gap by training and evaluating eight classifiers across three independent datasets, each subjected to a distinct adversarial condition that simulates a realistic data challenge. Beyond accuracy, we assess how each model responds to these challenges, identify the clinical features driving predictions, and evaluate whether improvement techniques— hyperparameter tuning, feature selection, and ensemble learning—deliver consistent gains or context-dependent ones.

2 Related Work

The application of machine learning to preeclampsia prediction has expanded rapidly over the past several years. Aziz et al. (2023) were among the earlier contributors, comparing logistic regression, decision trees, and random forests on clinical data and demonstrating that ensemble methods outperform single-model approaches, though their findings were limited by a small and homogeneous dataset. Ghosh et al. (2024) extended this direction by integrating both physiological and demographic features, achieving improved recall using logistic regression, SVM, and random forests, but their evaluation did not address model robustness or class imbalance.

At a larger scale, PIERS-ML Study Group (2024) applied boosting and ensemble techniques to a multi-country dataset of over 8,800 patients and reported strong AUC and sensitivity values; however, the complexity of the resulting models raised concerns about clinical interpretability. Systematic reviews by Sufriyana et al. (2024) and Alhamzawi et al. (2024) highlighted a recurring problem in the field: inconsistent preprocessing protocols and evaluation strategies across studies make direct comparison difficult, and challenges such as class imbalance and noisy data remain inadequately addressed. More recently, Li et al. (2025) demonstrated that incorporating inflammatory biomarkers alongside clinical data measurably improves predictive accuracy using feature engineering and data integration, and a meta-analysis by Chen et al. (2026) confirmed that while ML models consistently achieve high performance on benchmark data, reproducibility and generalisation across populations remain open problems.

The literature as a whole points toward a clear need for studies that use diverse data sources, apply consistent preprocessing, and rigorously test models under realistic rather than idealised conditions. This study directly addresses that need.

3 Data Description and Preprocessing

Three publicly available datasets were used, each processed and evaluated independently to reflect the diversity of real-world maternal health data.

Dataset 1 – Maternal Health Risk This dataset contains 1,014 records collected from IoT-based maternal monitoring systems in Bangladeshi hospitals. Features include age, systolic and diastolic blood pressure, blood sugar (BS), body temperature, and heart rate. The prediction target is a three-class risk label (low, mid, high). The data was complete with no missing values, though a mild class imbalance existed: low-risk cases were most frequent (406 records), followed by mid-risk (336) and high-risk (272). Correlation analysis revealed a strong relationship between systolic and diastolic blood pressure, while other features showed relatively weak intercorrelations, indicating largely independent contributions to the prediction task.

Dataset 2 – Factors for Preeclampsia This dataset contains 400 records with 25 features covering demographic, physiological, and biochemical markers including TSH, PLGF:sFLT ratio, BMI, creatinine, and glycerides. The binary target indicates hypertension status. No missing values were present, and the two classes were approximately balanced. Several biochemical features exhibited high skewness and were log-transformed before modelling to reduce the influence of extreme values.

Dataset 3 – Preeclampsia in Pregnant Women This dataset was provided as separate training and test files (162 and 41 records respectively) with 17 clinical and obstetric features. Three columns were entirely missing and were removed. One remaining missing value was imputed using the training-set median. Notably, the training file contained 87 duplicate rows, which were removed, leaving 75 unique records for training—making this the most challenging dataset due to its limited size.

Preprocessing Pipeline

Each dataset was processed through a consistent pipeline: removal of fully-missing columns and duplicate rows; median imputation for residual missing values; label encoding of categorical targets using `LabelEncoder`; `StandardScaler` normalisation fitted on training data only and applied to validation and test sets to prevent data leakage; and stratified train/validation/test splits of 64%/16%/20% (648/163/203 records) for Datasets 1 and 2, with the pre-defined split used for Dataset 3. EDA visualisations including histograms, boxplots, and correlation heatmaps were produced for each dataset to guide preprocessing decisions.

4 Methodology

Models

Eight supervised classifiers were implemented and evaluated: Logistic Regression (linear baseline), Decision Tree, Random Forest (bagging ensemble), SVM, KNN ($k = 5$), Gaussian Naive Bayes, Gradient Boosting, and XGBoost. XGBoost was excluded from Dataset 3 due to a feature-naming compatibility issue. All models were trained and evaluated independently on each dataset without cross-dataset merging.

Adversarial Conditions

A distinct adversarial condition was applied to each dataset to test robustness under realistic data challenges.

- **Dataset 1 – Minority Class Suppression (SMOTE):** SMOTE was applied to the training set, oversampling the minority classes to balance all three risk groups to 260 samples each (780 total). SMOTE was applied only to training data to prevent leakage into validation or test sets.
- **Dataset 2 – Label Noise:** Label noise was injected by randomly flipping 5% of training labels, simulating annotation uncertainty common in clinical settings where records may be mislabelled due to human error.
- **Dataset 3 – Distribution Shift (Feature Drift):** Gaussian noise was added to Systolic BP values in the test set, simulating the distribution shift that occurs when a model trained in one clinical environment is deployed in another with different measurement characteristics.

Model Improvement

Following baseline evaluation, each dataset underwent a model improvement phase:

- **Hyperparameter tuning** via `GridSearchCV`
- **Class-weight adjustment** (`class_weight='balanced'`) applied to tree-based models to address persistent imbalance.
- **Feature selection** guided by tree-based feature importance scores, removing the lowest-importance feature (body temperature in Dataset 1).
- **Soft-voting ensemble** construction combining the top-performing classifiers per dataset to leverage model diversity.

Evaluation

Performance was assessed using weighted F1-score, accuracy, and AUC-ROC. Five-fold stratified cross-validation assessed training stability. Final comparisons were made on held-out test sets to confirm generalisation.

5 Experimental Results and Discussion

Dataset 1: Maternal Health Risk

Under balanced conditions after SMOTE, tree-based models dominated. Random Forest achieved the best test performance with 84.7% accuracy, F1 of 0.848, and AUC of 0.951. Decision Tree led on validation but overfitted and fell to third on the test set, illustrating the risk of selecting models on validation performance alone. Feature importance analysis identified blood sugar (BS) as the strongest predictor, followed by systolic blood pressure. Body temperature had near-zero importance and was removed during feature selection, resulting in a marginal improvement. Table 1 summarises results for the top models, and Table 2 shows the direct impact of SMOTE on Random Forest performance.

Model	Val F1	Test F1	AUC
Random Forest	0.833	**0.848**	**0.951**
XGBoost	0.797	0.835	0.952
Decision Tree	0.839	0.815	0.880
Gradient Boost.	0.790	0.790	0.928
Logistic Reg.	0.604	0.604	0.778

Table 1: Dataset 1 results . Best test values in bold.

Dataset 2: Factors for Preeclampsia

Label noise caused a substantial performance drop across all models and reversed typical

Condition	Accuracy	F1
Without SMOTE	0.801	0.798
With SMOTE	**0.847**	**0.848**

Table 2: Effect of SMOTE on Random Forest, Dataset 1.

model rankings. Naive Bayes proved the most resilient (validation F1 = 0.596) while ensemble methods such as Random Forest suffered the most (F1 = 0.484). This reflects a fundamental property: high-capacity models that fit training data tightly will also memorise label errors, whereas simpler probabilistic models are less sensitive to individual corrupted labels. Five-fold cross-validation on Dataset 2 showed that Logistic Regression achieved the highest CV F1 mean (0.537), further confirming that simpler models are more stable under noisy conditions. Gradient Boosting with hyperparameter tuning and SMOTE achieved partial recovery, delivering the best test F1 of 0.549. Tables 3 and 4 summarise these results.

Model	Val F1	Test F1	AUC
Naive Bayes	**0.596**	0.466	0.483
XGBoost	0.575	0.522	0.540
Gradient Boost.	0.558	**0.549**	**0.538**
Random Forest	0.484	0.484	0.554
Logistic Reg.	0.463	0.463	0.431

Table 3: Dataset 2 results under label noise. Best per column in bold.

Condition	Accuracy	F1
Without label noise	0.725	0.718
With 5% label noise	0.563	0.558
After tuning + SMOTE	**0.609**	**0.610**

Table 4: Label noise impact and partial recovery via Gradient Boosting, Dataset 2.

Dataset 3: Preeclampsia in Pregnant Women

Despite feature drift on Systolic BP, the top three models converged to nearly identical test accuracy of approximately 82.9%. This resilience is notable given the dataset's small training size (75 unique records). The soft-voting ensemble combining Random Forest, Naive Bayes, and Gradient Boosting achieved the best validation F1 of 0.735, outperforming all individual models on the validation set. Naive Bayes reached the highest AUC of 0.951 on the test set. Hyperparameter tuning and feature selection did not improve results on this dataset, confirming that with limited data most features carry useful information and further model complexity is counterproductive. Table 5 presents the full results.

Model	Val F1	Test F1	AUC
Naive Bayes	**0.675**	0.829	**0.951**
Random Forest	0.602	**0.830**	0.919
Gradient Boost.	0.600	0.828	0.920
Decision Tree	0.591	0.591	0.698

Table 5: Dataset 3 results under Systolic BP feature drift. Best per column in bold.

Condition	Accuracy	F1
Without drift	0.878	0.872
With Systolic BP drift	0.829	0.830
Ensemble (RF + NB + GB)	0.733*	0.735*

Table 6: Feature drift impact and ensemble recovery, Dataset 3. *Validation set.

Cross-Dataset Discussion

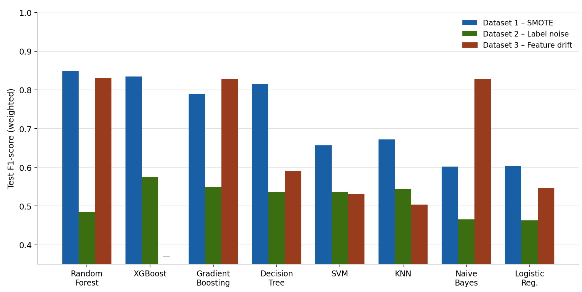

Figure 1: Test F1-score comparison across all three datasets. Blue = Dataset 1 (SMOTE), Green = Dataset 2 (label noise), Red = Dataset 3 (feature drift). XGBoost has no bar for Dataset 3 due to a feature-naming incompatibility. Dataset 2 scores are consistently the lowest across all models—a direct consequence of label noise degrading every classifier's ability to learn reliable decision boundaries.

Several consistent patterns emerge from the cross-dataset analysis. First, ensemble methods—Random Forest, Gradient Boosting, XGBoost—lead under clean or balanced conditions (Dataset 1), but this advantage disappears under label noise (Dataset 2), where Naive Bayes proves the most robust. This demonstrates that high-capacity models memorise corrupted labels, whereas simpler probabilistic models remain stable.

Second, SMOTE reliably improved performance when a class was clinically underrepresented. The gain on Dataset 1 came specifically from improved recall on the high-risk class—the most clinically critical group, where a missed detection has far greater consequences than a false alarm.

Third, feature selection benefited only Dataset 1 (where body temperature contributed near-zero variance), while in Datasets 2 and 3 all features carried useful signal and removing any of them consistently reduced performance. This highlights that feature selection decisions must be grounded in data-specific importance analysis rather than applied uniformly.

Fourth, hyperparameter tuning via GridSearchCV had mixed results: it improved XGBoost on Dataset 1 and Gradient Boosting on Dataset 2, but was negligible or slightly harmful for Random Forest and Naive Bayes across datasets. This reinforces that tuning must be validated on held-out data and cannot be assumed beneficial.

Finally, the ensemble in Dataset 3 succeeded specifically because it combined diverse model types—a bagging ensemble (RF), a probabilistic model (NB), and a boosting ensemble (GB)—producing genuinely complementary predictions that are more stable under distribution shift. Combining two tree-based learners, by contrast, adds limited diversity and provides little benefit.

6 Conclusion and Future Work

This paper presented a multi-source machine learning framework for early preeclampsia prediction evaluated under both standard and adversarial conditions across three independent datasets. By deliberately introducing class imbalance, label noise, and feature distribution drift as controlled challenges, the study provided a more realistic assessment of model behaviour than benchmark-only evaluation can offer.

Random Forest was the most consistently reliable classifier, achieving 84.7% test accuracy and AUC of 0.951 on Dataset 1 and remaining among the top performers on Dataset 3. It was therefore selected as the final recommended model for this project, owing to its stable performance across varied data conditions. Gradient Boosting with SMOTE delivered the best results under label noise on Dataset 2. A three-model ensemble achieved the strongest validation performance on the smallest, most difficult dataset.

The relationship between model complexity and performance proved non-trivial: complexity is an asset under clean or balanced conditions and a liability under noisy ones. No single optimisation technique universally improved outcomes—each has conditions under which it helps and conditions under which it adds little. The practical implication for clinical deployment is that a model must be tested under data conditions that reflect its intended environment, not just clean benchmark splits.

Future directions include applying temporal deep learning models to longitudinal biomarker sequences to capture the evolving nature of preeclampsia risk during pregnancy, using SHAP and LIME for per-prediction clinical reasoning to improve interpretability, and exploring federated learning for privacy-preserving multi-institutional collaboration that would allow models to be trained on larger and more diverse patient populations without sharing sensitive data.

Known Limitations

Several limitations should be considered. First, all three datasets are relatively small by machine learning standards—75 unique training records in Dataset 3 creates high variance in model evaluation, and results on small splits should be interpreted with caution. Second, the datasets are sourced from public repositories and may not represent the demographic diversity of global maternal populations, particularly non-Bangladeshi and non-South-Asian patient groups; models trained on this data may not generalise to populations with different risk factor distributions. Third, the label noise and feature drift conditions applied in this study are simplified simulations of real-world challenges— actual clinical data corruption is more complex, structured, and often systematic rather than random. Fourth, no external validation on a fully independent clinical cohort was performed; all evaluation is on held-out splits from the same source distributions. Finally, the biochemical features in Dataset 2 (e.g., PLGF:sFLT, TSH) may not be routinely available in low-resource clinical settings, limiting the practical applicability of models trained on them.

Authorship Statement

This project was completed as a team across five phases. Dana Al Mounayer led the data preprocessing pipeline, adversarial condition design, and the model improvement phase for Dataset 1. Sedra Al Jundi was responsible for baseline model implementation, cross-validation analysis, and the model improvement phase for Dataset 2. Jwdee Al amam led exploratory data analysis, feature importance interpretation, and the model improvement phase for Dataset 3. The final report, figures, and presentation were written and produced collaboratively by all three authors. All team members contributed to result interpretation and discussion. This project was completed entirely within the course team with no external collaborators.

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